

HR Analytics Employee Attrition Analysis Using Microsoft Excel

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Executive Summary

This report presents an HR Analytics project completed in Microsoft Excel. The project analyzes employee attrition using PivotTables, PivotCharts, Excel formulas, and an interactive dashboard to identify workforce trends and support business decision-making.

Introduction

Employee attrition impacts recruitment costs, productivity, and organizational stability. This project examines employee data to identify key drivers of attrition and visualize the findings through an interactive dashboard.

Objectives

- Analyze attrition across departments.
- Identify factors influencing employee turnover.
- Evaluate job involvement, stock options, training, and job satisfaction.
- Build an interactive Excel dashboard.

Dataset Overview

The dataset contains 1,470 employee records from the Human Resources domain, including demographic, employment, compensation, performance, and attrition information.

Data Preparation

The dataset was checked for completeness, reviewed for missing values and duplicates, formatted as an Excel Table, validated for data consistency, and prepared for PivotTable analysis.

Methodology

Excel Tables, PivotTables, PivotCharts, COUNTIF, AVERAGE, CORREL, percentage calculations, slicers, and dashboard design techniques were used. The project included basic analysis, medium analysis, and dashboard development.

Dashboard Overview

The dashboard contains five KPI cards, six business charts, and interactive slicers for Department, Gender, Marital Status, Education, and OverTime.

Key Findings

Research & Development recorded the highest attrition. Single employees showed the highest turnover. Lower job involvement and lack of stock options were associated with higher attrition. Employees with more training generally had lower attrition. Job satisfaction varied only slightly across job roles.

Recommendations

Improve employee engagement, strengthen training programs, review compensation and stock-option policies, focus retention efforts on high-risk employee groups, and monitor workforce trends regularly.

Conclusion

The project demonstrates how Microsoft Excel can transform raw HR data into meaningful business insights through structured analysis and interactive visualization.

Project Summary

Metric	Value
Total Employees	1,470
Employees Left	237
Attrition Rate	16.12%
Charts	6
Slicers	5